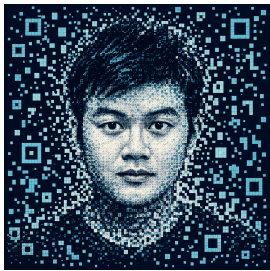


Bingru Huang

Curriculum Vitae



Assistant Researcher
School of Mathematical Sciences
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Academic Appointments

Assistant Researcher	Current
School of Mathematical Sciences, University of Science and Technology of China	
Postdoctoral Researcher	2023–2025
University of Science and Technology of China	

Education

Ph.D. in Computational Mathematics	September 2017–June 2023
University of Science and Technology of China School of Mathematical Sciences; Advisor: Prof. Falai Chen	
B.Sc. in Information and Computational Science	September 2013–June 2017
Wuhan University of Technology	

Research Interests

Mathematical foundations of splines and their applications in geometric design and numerical analysis:

- Isogeometric analysis;
- Approximation theory;
- Computer-aided geometric design;
- Mathematical theory of adaptive splines.

Teaching

Theory of Multivariate Function Approximation	Spring 2025
Graduate course; Instructor	
Spline Functions and Approximation Theory	Fall 2024; Fall 2025
Graduate course; Instructor	
Linear Algebra B1	Spring 2022
Teaching Assistant	
Mathematical Analysis B1 & B2	2019–2021
Teaching Assistant	

Publications

8. **Bingru Huang** and Falai Chen, “A preliminary study on the dimensional stability classification of polynomial spline spaces over T-meshes,” *Computer Aided Geometric Design*, 130, 102609, 2026. DOI: 10.1016/j.cagd.2026.102609.
7. Shicong Zhong, **Bingru Huang**, and Falai Chen, “Modified HB-splines,” *Journal of Computational and Applied Mathematics*, 485, 117467, 2026. DOI: 10.1016/j.cam.2026.117467.
6. **Bingru Huang** and Falai Chen, “Dimension of bi-degree (d, d) spline spaces with the highest order of smoothness over hierarchical T-meshes,” *Science China Mathematics*, 69, 2195–2218, 2026. DOI: 10.1007/s11425-023-2499-9. arXiv: 2507.11047.
5. Shicong Zhong, **Bingru Huang**, and Falai Chen, “Basis construction for polynomial spline spaces over arbitrary T-meshes,” *Computer Aided Geometric Design*, 126, 102534, 2026. DOI: 10.1016/j.cagd.2026.102534. arXiv: 2508.12950.
4. Shicong Zhong, **Bingru Huang**, and Falai Chen, “Weighted Hierarchical B-Spline Basis,” *Journal of Computer-Aided Design & Computer Graphics*, 38(4), 731–741, 2026. [In Chinese] DOI: 10.3724/SP.J.1089.2024-00337.
3. **Bingru Huang** and Falai Chen, “An Algorithm to Compute the μ -Bases of Rational Parametric Surfaces with Respect to One Variable,” *Communications in Mathematics and Statistics*, 12(3), 523–541, 2024. DOI: 10.1007/s40304-022-00300-y.
2. **Bingru Huang** and Falai Chen, “On the stability of the dimensions of spline spaces with highest order of smoothness over T-meshes,” *Journal of Computational and Applied Mathematics*, 441, 115681, 2024. DOI: 10.1016/j.cam.2023.115681.
1. **Bingru Huang** and Falai Chen, “Computing μ -Bases of Univariate Polynomial Matrices Using Polynomial Matrix Factorization,” *Journal of Systems Science and Complexity*, 34(3), 1189–1206, 2021. DOI: 10.1007/s11424-020-9314-6.